



# BAAZ JKBOSE

🔥 2026 CLASS 11<sup>th</sup> 🔥

**Cell:  
The Fundamental Unit of Life**

**ZOOLOGY LECTURE-2**

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# TOPICS TO BE COVERED

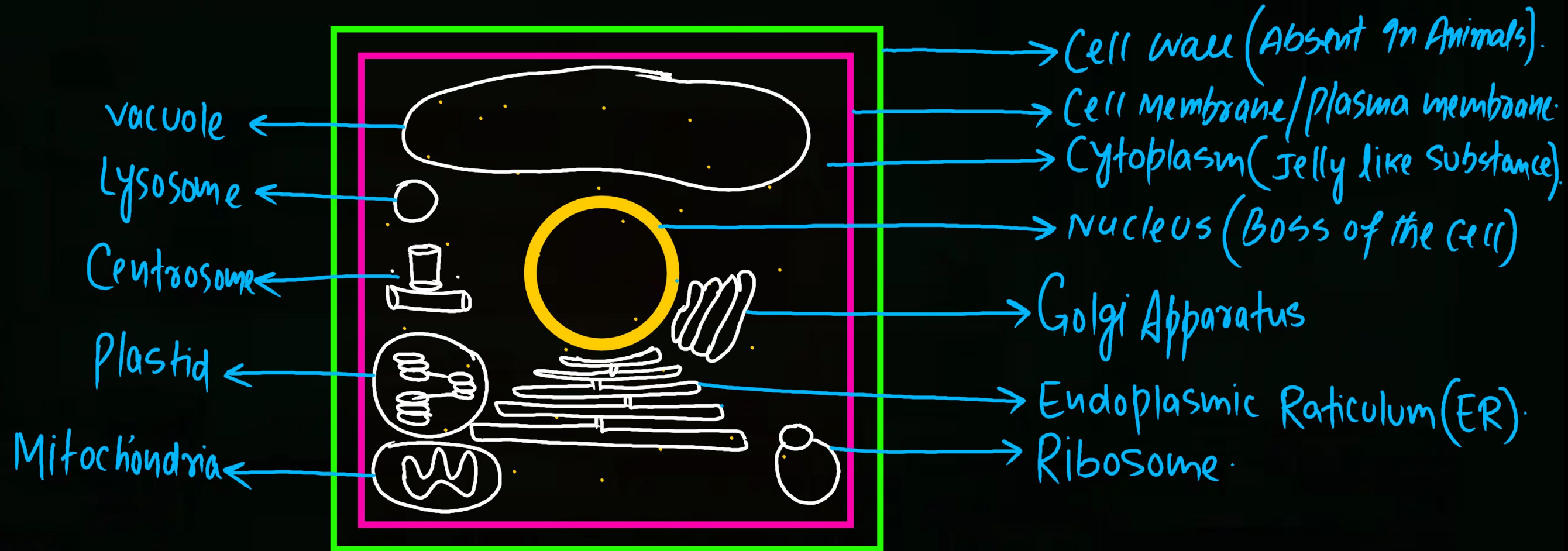
**1** STRUCTURE OF A CELL

**2** CELL MEMBRANE

**3** CELL WALL

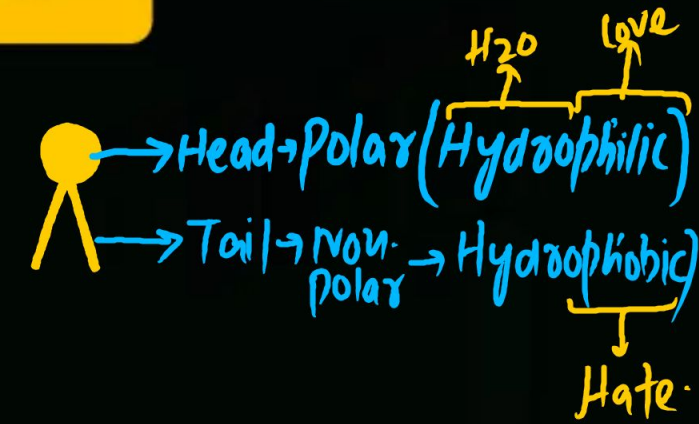
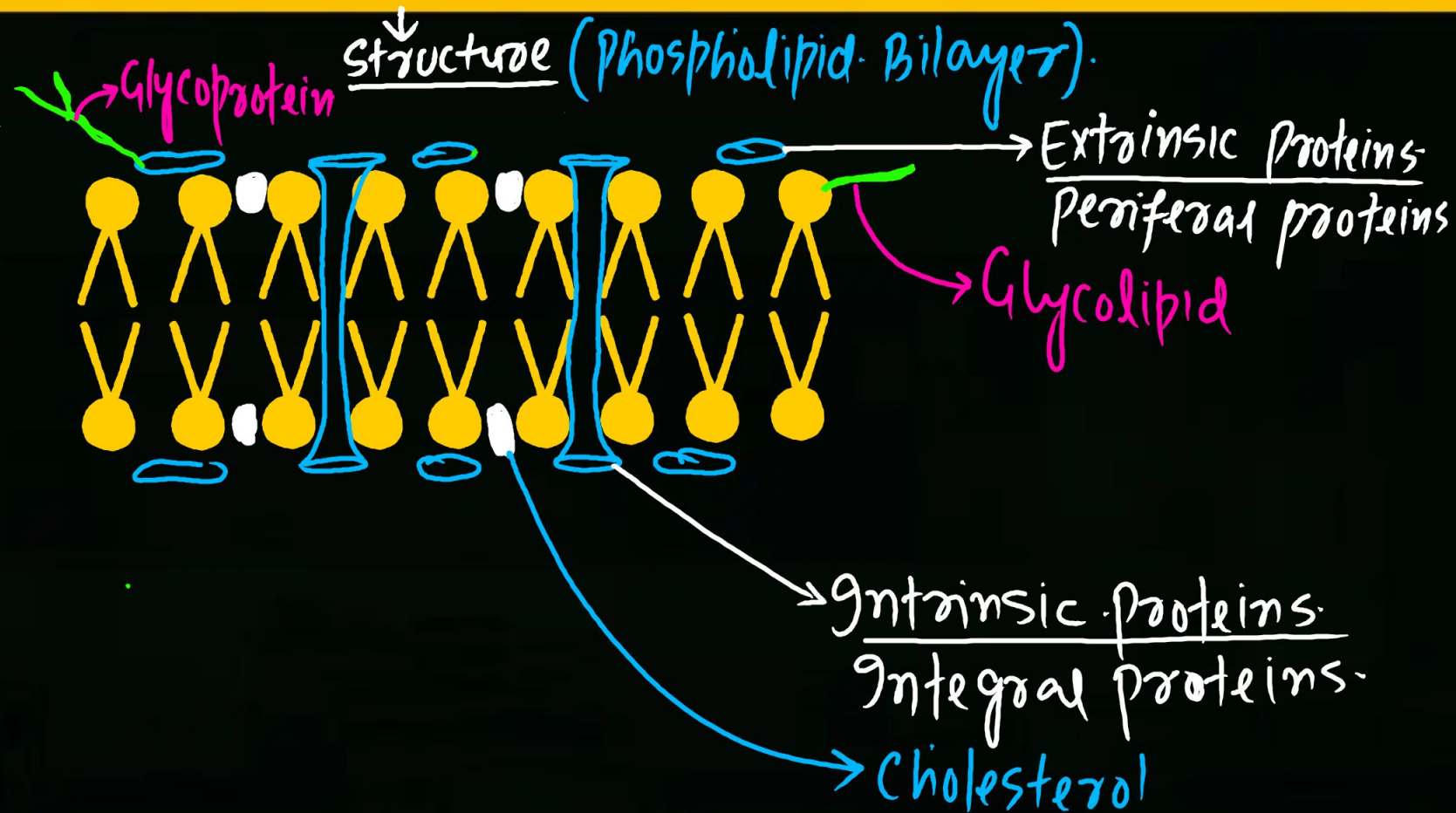
**4** PREVIOUS NEET QUESTIONS

# STRUCTURE OF A CELL



# Cell Membrane or Plasma Membrane

- Composed of
- ① Lipid (Phospholipid) ✓
  - ② Protein ✓
  - ③ Carbohydrate ✓
  - ④ Water ✓
  - ⑤ Cholesterol ✓



# Cell Membrane or Plasma Membrane

- It surrounds the cytoplasm of all cells and separates the cell's interior from the external environment.
- Its detailed structure was understood after the invention of the electron microscope (1950s).
- Chemical studies (especially in human RBCs) showed that the membrane is mainly made of lipids and proteins, with carbohydrates also present.
- The major lipids are phospholipids, arranged in a bilayer, along with cholesterol.
- Phospholipids have polar (hydrophilic) heads facing outward and non-polar (hydrophobic) tails facing inward.
- The protein-lipid ratio varies among cells; in human erythrocytes, it is about 52% protein and 40% lipid.

# Cell Membrane or Plasma Membrane

- Membrane proteins are of two types:
  - (i). **Peripheral proteins**:- present on the membrane surface.
  - (ii). **Integral proteins**:- partially or completely embedded in the membrane.

Singer and Nicolson (1972) proposed the Fluid Mosaic Model, which is widely accepted. According to this model, the lipid layer is quasi-fluid, allowing lateral movement of proteins (membrane fluidity).

Membrane fluidity is essential for functions like cell growth, intercellular junction formation, secretion, endocytosis, and cell division.

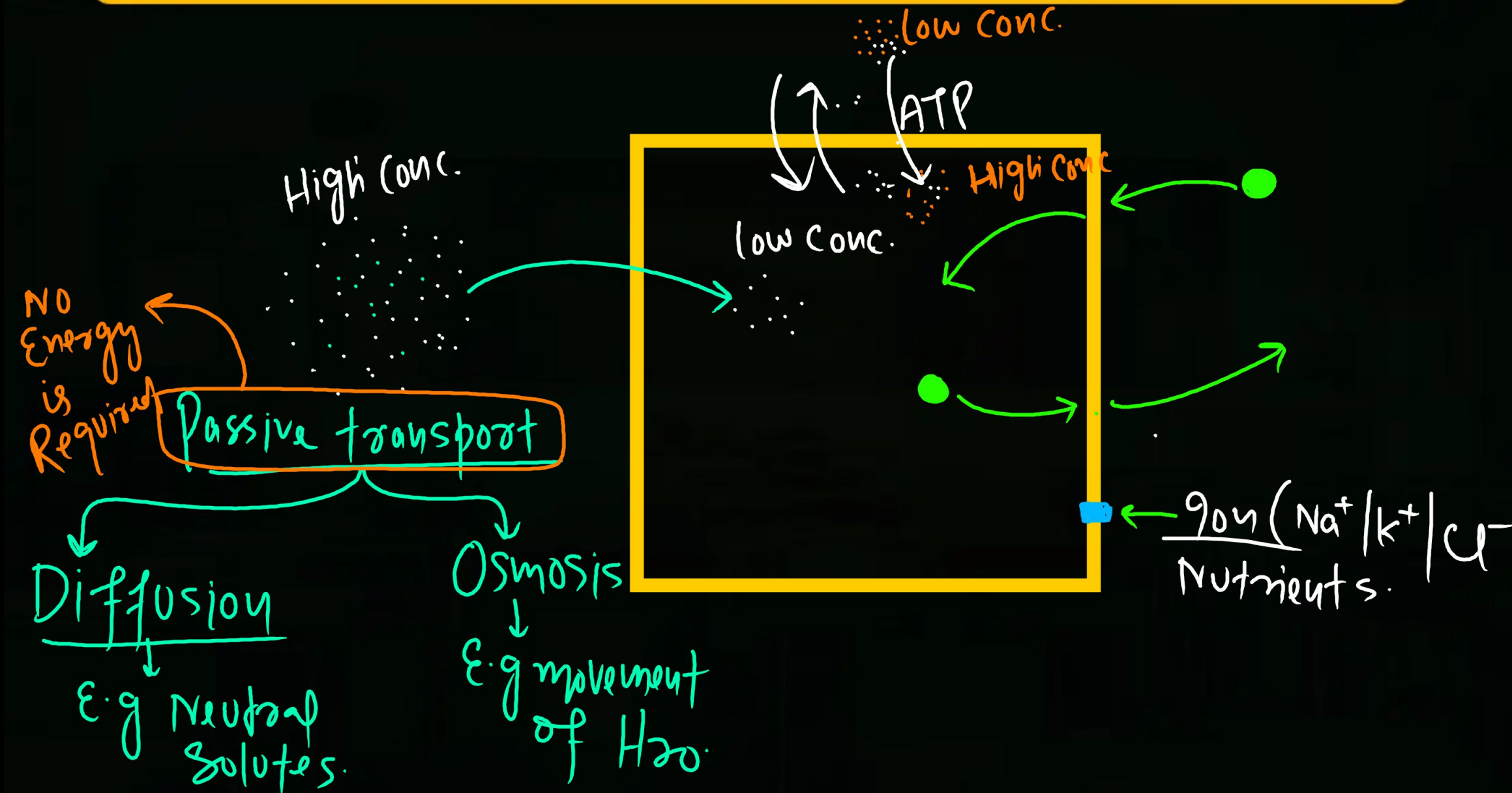
# Question: NEET-2010

The plasma membrane consists mainly of-

- 1 Phospholipids embedded in a protein bilayer
- 2 Proteins embedded in a phospholipid bilayer. ✓
- 3 Proteins embedded in a polymer of glucose molecules
- 4 Proteins embedded in a carbohydrate bilayer.



# Functions of the Cell Membrane



Active transport  
 ↓  
 Energy (ATP) is Required  
 E.g. Na<sup>+</sup>/K<sup>+</sup>

# Functions of the Cell Membrane

- **Selective Permeability**: The membrane controls the movement of substances into and out of the cell.
- **Transport**: The membrane contains proteins that facilitate the transport of ions, nutrients, and waste products. This includes:
  - (i). **Passive transport**:- That does not require energy.
    - ✓(a). **Simple diffusion**:- Neutral solutes may move across the membrane along the concentration gradient.
    - ✓(b). **Osmosis**:- Movement of water by diffusion is called osmosis.
  - (ii). **Active transport**:- Polar molecules cannot cross the non-polar lipid bilayer directly. They require carrier proteins present in the membrane. In active transport, substances move against the concentration gradient (from lower to higher concentration). This process requires energy in the form of ATP. Example:  $\text{Na}^+/\text{K}^+$  pump.

## Question: NEET-2010

Select the correct statement from the following regarding cell membrane.

1 Na<sup>+</sup> and K<sup>+</sup> ions move across cell membrane by passive transport X

2 Proteins make up 60 to 70% of the cell membrane. ✓

3 Lipids are arranged in a bilayer with polar heads towards the inner part. X

4 Nucleus functions as control centre of cell X

## Question: NEET-2008

H/W.

Keeping in view the 'fluid mosaic model' for the structure of cell membrane, which one of the following statement is correct with respect to the movements of lipids and proteins from one lipid monolayer to the other (described as flip-flop movement)?

1 While proteins can flip-flop, lipids can not

2 Neither lipids, nor proteins can flip-flop

3 Both lipids and proteins can flip-flop

4 While lipids can rarely flip-flop, proteins cannot.



## Question: NEET-2007

Which one of the following is not a constituent of cell membrane?

1 Glycolipids

2 Proline.

3 Phospholipids

4 Cholesterol



## Question: NEET-2005

H/W.

According to widely accepted "fluid mosaic model" cell membranes are semi-fluid, where lipids and integral proteins can diffuse randomly. In recent years, this model has been modified in several respects. In this regard, which of the following statements is incorrect?

- 1 Proteins in cell membranes can travel within the lipid bilayer.
- 2 Proteins can also undergo flip-flop movements in the lipid bilayer
- 3 Proteins can remain confined within certain domains of the membrane
- 4 Many proteins remain completely embedded within the lipid bilayer

**Thank**  
**You**